



SEA-BIRD
SCIENTIFIC

User manual

ECO Triplet-w

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Section 1 ECO™ quick start guide

⚠ WARNING



fDOM sensors use an ultraviolet LED light source. Do not look directly at a UV LED when it is on. It can cause damage to the eyes. Keep products that have UV LEDs away from children, pets, and other living organisms. Wear polycarbonate UV-resistant safety glasses to protect the eyes when a UV LED is on.

This quick start guide gives the steps necessary to set up and make sure that the ECO sensor operates and collects data before it is deployed.

What's in the box—

- ECO sensor
 - Six 9V AA-size lithium batteries (model-specific, shipped separately)
 - blue-tipped power connector and lock collar for sensors with internal batteries
 - dummy plug and lock collar
 - model-specific spare parts kit
 - model-specific fluorescent stick
 - CD or USB drive with software, user manual, and calibration or characterization information.
1. If the sensor is so-equipped, install the manufacturer-supplied batteries:
 - a. Remove the end flange of the sensor.
 - b. Disconnect the battery pack and remove it from the sensor.
 - c. Install new batteries.
 - d. Connect the battery pack to the sensor again and install the end flange again.
 2. Install the manufacturer-supplied software on a PC.
 3. Verify that the sensor operates and collects data:
 - a. Connect the sensor to the PC and start the software.
 - b. If necessary, supply external power to the sensor (models with no internal batteries).
 - c. Remove and replace the protective cover on the optical face several times, or use the fluorescence stick that shipped with the sensor, to change the value of the collected data.
 - d. Push **Stop Data** to stop data collection.
 4. Verify that the sensor transmits data:
 - Select the *Transfer Data* tab in the software.
 - Push **Receive Data**.
 - Select a location on the PC in which to save the data, and enter a file name.
 - Push **Save**.
 5. If necessary, attach the sensor to the frame on which it will be deployed.
 6. Make sure to do maintenance on the sensor at regular intervals.

Section 2 Specifications

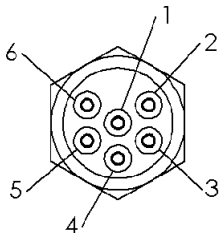
ECO sensors measure different parameters of the Earth's natural waters. The sensors are available in several models and with a variety of optional features.

2.1 Mechanical

	Wiper (W)	Wiper and Battery (WB)
Diameter	8.08 cm	
Length	22.1 cm	33.34 cm
Depth rating	1500 m	1000 m
Temperature range	0–30 °C	
Weight in air, water	1.25 kg, 0.29 kg	2.1 kg, 0.43 kg

2.1.1 Bulkhead connectors

2.1.1.1 6-contact connector

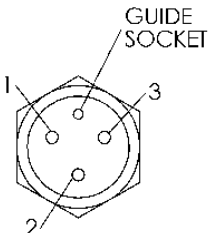
Contact	Function	MCBH-6-MP
1	Ground	
2	RS232 RX	
3	Reserved	
4	Voltage in	
5	RS232 TX	
6	Analog 1	

Note: Contact 3 on two-parameter sensors is a second analog output.

2.1.1.2 Battery

Additional bulkhead connector on sensors with internal batteries.

Use the three-contact blue-tipped power connector that is supplied with the sensor to supply power to the sensor.

Contact	Function	MCBH-3-FS connector
1	Voltage in	
2	No connect	
3	Battery out	

2.2 Electrical

Input	7–15 VDC
Current draw, typical	60 mA
Current draw, standby	140 µA
Current draw, active wiper	200 mA
Linearity	99%

Specifications

2.3 Communications

Sample rate	to 4 Hz
Data storage	67000 samples
RS232 output rate	19200 baud
Output resolution	12 bit
Digital output maximum	4130 ±30 counts

2.4 Optical

Note: The manufacturer has changed the nomenclature for the output of all dissolved organic matter fluorometers. Fluorescent dissolved organic matter (FDOM) replaces colored dissolved organic matter (CDOM). FDOM aligns the fluorescence measurement with the output description.

2.4.1 Three-parameter fluorometer and scattering

Parameter	Wavelength EX/EM	Range, Sensitivity
Chlorophyll (Chl)	470/695 nm	0–30, 0.015 µg/L
		0–50, 0.025 µg/L
Fluorescent Dissolved Organic Matter (fDOM)	370/460 nm	0–375, 0.28 ppb
Uranine (UR)	470/530 nm	0–300, 0.15 ppb
Phycocyanin (PC)	630/680 nm	0–175, 0.086 ppb
Phycoerythrin (PE), Rhodamine (Rh)	530/595 nm	0–175, 0.086 ppb

Parameter	Wavelength	Range, Sensitivity
Scattering	412 nm, 470 nm, 532 nm, 650 nm, 880 nm	0–5, 0.003 m ⁻¹
	700 nm	0–3, 0.002 m ⁻¹
		0–5, 0.003 m ⁻¹

Section 3 Operation and maintenance

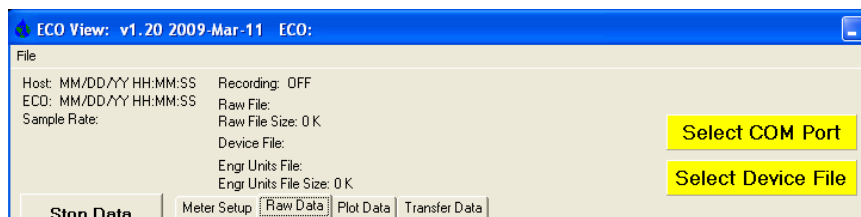
3.1 Verify sensor operation

⚠ WARNING	
	fDOM sensors use an ultraviolet LED light source. Do not look directly at a UV LED when it is on. It can cause damage to the eyes. Keep products that have UV LEDs away from children, pets, and other living organisms. Wear polycarbonate UV-resistant safety glasses to protect the eyes when a UV LED is on.

NOTICE
Do not supply more than 15 VDC to the sensor. More than 15 VDC will damage the wiper.

Make sure that the sensor operates before further setup and deployment.

1. Connect the 6-contact connector on the optional test cable (refer to the section on the test cable for details) to the sensor.
2. Remove the cap that protects the optical face of the sensor.
3. Connect a serial-to-USB adapter to the test cable to connect the test cable to the PC.
4. Connect the sensor to a power supply:
 - Connect sensors with internal batteries to the manufacturer-supplied, three-contact, blue-tipped power connector. The sensor comes on.
 - Connect sensors without internal batteries to the optional test cable and a regulated power supply set at 12 VDC.
5. Start the software from the manufacturer-supplied CD.
 - a. Select the COM port on the PC.
 - b. Select the device file for the sensor from the CD.
 - c. Select the baud rate if necessary. The default is 19200.



6. Turn on the power supply.
The sensor comes on.
7. Push **Start Data**.
8. Go to the *Raw Data* tab of the software.
The data collected by the sensor shows in the "Signal" column.

Figure 1 Format of data collected by most ECO sensors

Meter Setup	Raw Data	Plot Data	Transfer Data			
06/14/12 06:06:05	532	267	660	3070	695	78
06/14/12 06:06:06	532	315	660	3406	695	89
06/14/12 06:06:07	532	437	660	3861	695	127
06/14/12 06:06:08	532	509	660	4122	695	175
06/14/12 06:06:09	532	2577	660	4122	695	629
06/14/12 06:06:10	532	4122	660	4122	695	957
06/14/12 06:06:11	532	4122	660	4122	695	970
06/14/12 06:06:12	532	4122	660	4122	695	868
06/14/12 06:06:14	532	4122	660	4122	695	946
Date	Time	Wave-length	Signal	Wave-length	Signal	Wave-length

Note that RT and puck sensors show 9's instead of the Date and the Time.

- Look at the maximum data value for the sensor. Hold a finger, the protective cap, or fluorescent stick if the sensor is a fluorometer, 1–4 cm away from the optical face of the sensor.

The data value in the "Signal" column in the *Raw Data* tab will increase toward the maximum data value specified for the sensor.

- Scattering and turbidity sensors: use a finger or the protective cap.
- FDOM and phycocyanin sensors: use the blue fluorescent stick.
- Chlorophyll or phycoerythrin sensors: use the orange fluorescent stick.
- Uranine sensors: use the yellow fluorescent stick.
- PAR sensors: point the sensor into the light.

- Push **Stop Data**.

The Bio-wiper closes on sensors so-equipped. If the power is turned off in mid-cycle, the Bio-wiper starts at the beginning of the cycle when power is supplied again.

3.2 Set up sensor for deployment

- Refer to the previous section to make sure that the sensor operates correctly.
- Replace the test cable with a sea cable for deployment.
- If necessary, remove the protective cap from the sensor.
- Use either the blue-tipped power connector (sensors with internal batteries only) or an external power supply to supply power to the sensor for deployment. If the power connector and a cable are both attached, the power is supplied by the equipment that supplies the highest voltage.
 - Attach the blue-tipped power connector to the three-contact connector. The sensor will begin operation as set up by the user. The manufacturer recommends this mode for moored applications.
 - Attach a sea cable to the six-contact connector and turn on the power supply (sensors without internal batteries).
- Refer to the section on [Additional operations](#) on page 11 for details on how to set up the sensor for a specific application.

3.3 Monitor data

Monitor the data from the sensor in counts. The number of columns of data will vary depending on whether the user has a one-, two-, or three-parameter sensor.

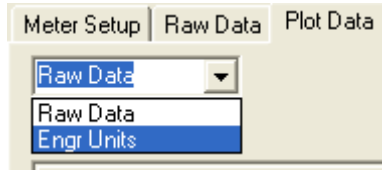
- Make sure the sensor has power supplied and is on.
- Push **Start Data**.
- Go to the *Raw Data* tab to see the data that is collected by the sensor.

Refer to the [Verify sensor operation](#) on page 7 to see the format of the data.

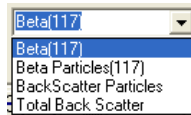
Note: RT and puck sensors usually show 9's as placeholders in the date and time columns.

3.3.1 Monitor data collected in engineering units

1. Go to the *Plot Data* tab.
2. Select "Engr Units" from the drop-down menu near the top of the tab.



3. Select the type of units to see.



The software calculates the engineering units to show in the *Plot Data* tab.

Note: Data is saved in counts, not engineering units.

3.4 Save data

Save the data collected by the sensor in the sensor's memory (if so equipped), on the PC, or both.

Note: RT and puck sensors do not store data. Save the data from these sensors in real-time to the PC or to a data logger.

3.4.1 Save data to sensor

1. Stop the sensor if it is on.
This puts the sensor in standby mode.
2. Push **Turn Logging ON/OFF**.

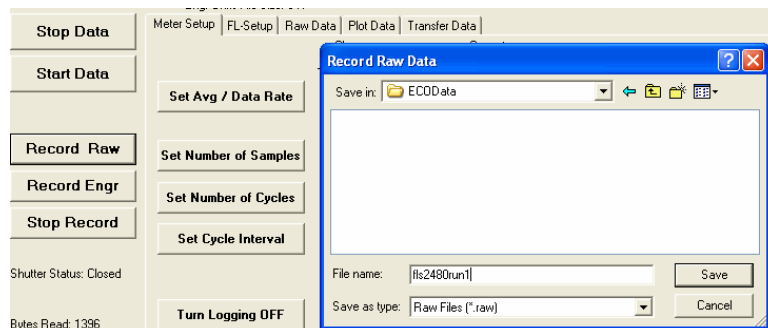


This turns on the logging, or data storage, in the sensor.

3. Push **Store to Flash**.
The yellow **Setup not Stored** warning will no longer show. The sensor will save data internally the next time the user turns it on.

3.4.2 Save data to PC

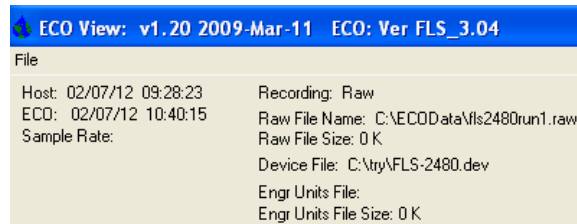
To save data to the PC in counts, push **Record Raw**. To save data in engineering units, push **Record Engr**. Data collected by the sensor is saved in real-time to the PC.



1. Push **Record Raw**.

The *Record Raw Data* window shows in the software.

2. Select a location on the PC to store the data.
3. Type a file name.
4. Push **Save**.
5. Push **Record Engr**.
The *Record Engineering Data* window shows in the software.
6. Select a location on the PC to store the data.
7. Type a file name.
8. Push **Save**.
9. Make sure the PC is configured to save data.
 - The file names from step 3 and step 7 will show in the software.



10. Push **Start Data**.
The software will show the PC file sizes.

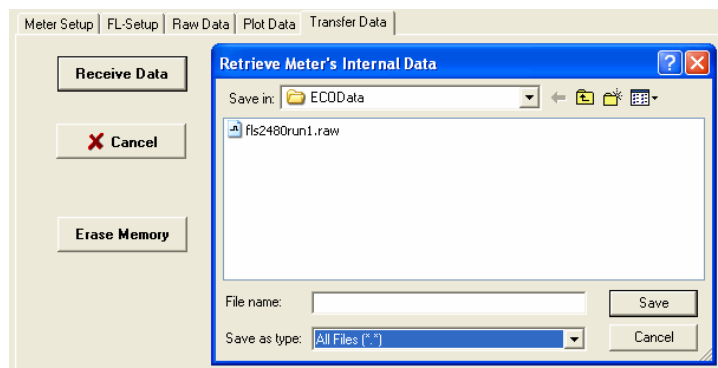
3.5 Get data from sensor

NOTICE

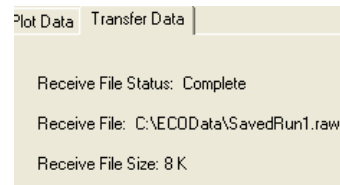
Use only the batteries recommended by the manufacturer as replacements. Do not mix chemistries, V-Ah, or new and used batteries.

Get the data from the sensor and save it to the PC.

1. Make sure that the sensor is not on but has power supplied to it.
2. Select the *Transfer Data* tab in the software.
3. Push **Receive Data**.



4. Select a location on the PC in which to save the data.
5. Type a file name in the **Retrieve Meter's Internal Data** window.
6. Push **Save**.
The software saves the data from the sensor to the PC.
7. Make sure that the data move is complete.



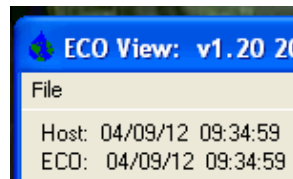
8. Open the data file to make sure that the data is on the PC.
Contact the manufacturer to get a spreadsheet template for an *ECO* sensor.
9. Push **Erase Memory** to erase the data from the memory of the sensor.

3.6 Additional operations

3.6.1 Set date and time

Make sure that the sensor is connected to a power supply that is turned on. Make sure that the software is open.

1. If the sensor is in operation, push **Stop Data** to stop the sensor.
2. Select the *Meter Setup* tab.
3. Push **Set Date and Time**.
The software adjusts the time shown by the sensor to agree with the PC.
4. Push **Get Date/Time/Setup** to make sure that the sensor and the PC show the same current time.



3.6.2 Adjust options for data collection

The manufacturer sets *ECO* sensors to operate at approximately 1 Hz, with the optional internal data storage turned on.

Table 1 Options to collect data

ECOView option	How it works
Set Avg/Data Rate	Adjust from 1 to 65535. Examples:
	1-parameter sensors—approximately 1 Hz = 65; approximately 2 Hz = 30
	2-parameter sensors—approximately 1 Hz = 30; approximately 2 Hz = 15
	3-parameter sensors—approximately 1 Hz = 18; approximately 2 Hz = 6
	PAR sensors—approximately 1 Hz = 310; approximately 2 Hz = 170
Set Number of Samples	Adjust from 0 to 65535. Use 0 to operate continuously.
The three options below apply only to sensors equipped with internal memory	
Set Number of Cycles	Adjust from 0 to 65535. Select the number of sample groups that the sensor will collect between low-power states.
Set Cycle Interval	Adjust the time interval between sample cycles. Do not type the colons. The minimum is 5 seconds.
Turn Logging ON	Push to turn data storage off or on (sensors with internal data storage only).

Adjust any of the options to collect data in the *Meter Setup* tab.

1 Enter the new value into the variable box.	2 Push the related button that is to the left of the variable box.	3 Push Store to Flash . The new value shows in the column <u>Current Ram Settings</u> .
--	--	--

In the example above, the 3-parameter sensor will operate at an "Average" of 18 and a "Data Rate" of 1.12 Hz. The sensor will collect data 10 rows of data for 3 cycles, with a low-power interval of 15 seconds after each cycle. The sensor stops after the 10th row of the third cycle of data is collected.

Table 2 Data collection examples

Collect moored data Set Avg/Data Rate = ± 1 Hz Set Number of Samples = 50 Set Number of Cycles = 24 Set Cycle Interval = 006000 Turn Logging ON/OFF = ON The sensor will collect data once per second, 50 times every 60 minutes for 24 hours, and store the collected data.	Collect profiling data Set Avg/Data Rate = ± 1 Hz Set Number of Samples = 0 Set Number of Cycles = N/A Set Cycle Interval = N/A Turn Logging ON/OFF = ON The sensor will collect data once per second and store collected data until the power is turned off.
---	--

If the sensor is set up to collect data intermittently, as for a moored deployment, it may be in a low-power condition. It is not possible to communicate with the sensor in this condition.

1. To start communication again, disconnect the sensor's power supply for one minute.
2. Connect the power again and push **Stop Data** several times.
3. Select the *Meter Setup* tab. Refer to [Adjust options for data collection](#) on page 11.
4. Type **0** into the **Number of Samples** variable box.
5. Push **Set Number of Samples**.
6. Push **Store to Flash**.
The sensor operates continuously.

Make sure that the sensor operates continuously.

1. Select the *Raw Data* tab.
2. Push **Start Data**.
3. Let the sensor operate for 10 samples or more.
4. Push **Stop Data**.

3.6.3 Adjust views in plot data tab

The *Plot Data* tab in the software lets the user see the data that the sensor has collected.



Button	Function	Description
1	Resume	Push to start or stop the data that shows.

2	Pause	The x-axis will stop.
3	Axes scroll	Move either axis up or down, or right or left.
4	Axes zoom	Move either axis up or down, or right or left.
5	Zoom out	Decrease the detail by 2x.
6	Zoom in	Increase the detail by 2x.
7	Zoom box	Draw a box around the data area to zoom all axes.
8	Cursor	Move the cursor bar to a certain data point.
9	Copy	Copy the current data plot to the PC clipboard.
10	Save	Save an image of the current data plot to the PC.
11	Print	Send a snapshot of the current data plot to a printer.

Enter the units of data to view from the drop-down menu above the black view area ($\mu\text{g/L}$, ppb, scattering, etc.)

3.7 General maintenance

NOTICE

Do not use abrasive cleaner on the optical face of the sensor. It will cause scratches on the optical epoxy and glass.

NOTICE

Do not use acetone or other solvents to clean any part of the sensor.

1. After each cast or exposure to natural water, flush the sensor with clean fresh water.
2. Use soapy water to clean any grease or oil on the optical face of the sensor. It is made of plastic and can be damaged if an abrasive cleaner is used.
3. Dry the sensor with a clean soft cloth.
4. Install the dummy plug and lock collar to protect the bulkhead connector.
5. Install the protective cap on the optical face.

3.7.1 Wiper replacement

Replace the sensor's wiper with a new one from the wiper replacement kit (sold separately).

1. Use a 1/16" hex wrench to loosen the wiper's set screw.

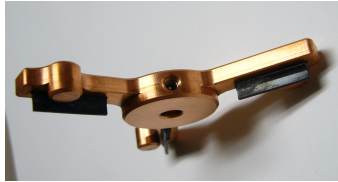
Figure 2 Wiper set screw loosened



2. Lift the wiper off of the wiper motor shaft.
3. Pull down on the long exposed part of the wiper to remove the rubber wiper from the copper wiper arm.



4. Push a new rubber wiper into the copper wiper arm. It may be necessary to move the rubber back and forth to get it completely through the slot.



5. Use a small Phillips screwdriver to remove the screws that attach the faceplate to the optics face.
6. Keep the screws.
7. Clean the Bio-wiper and faceplate with soapy water.
8. Rinse and dry thoroughly.
9. Use a Scotch-Brite® or similar pad to carefully buff the Bio-wiper, faceplate, and copper ring until they are shiny.
10. Use an isopropyl alcohol-saturated cotton swab to clean the Bio-wiper shaft and the shaft hole.
11. Make sure the Bio-wiper, faceplate, and copper ring are completely dry.
12. Attach the faceplate again.
13. Put the wiper arm onto the wiper arm shaft but do not tighten.
14. Put the wiper arm close to the optical face.
15. Make sure that the wiper is the correct height:
 - Make sure that the rubber lightly touches the copper faceplate.
 - The rubber will bend approximately 30 degrees against the optical face, which is higher than the copper faceplate.



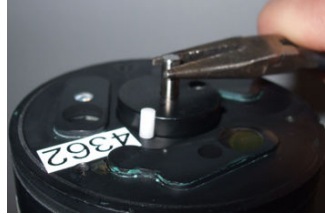
16. Use the appropriate hex key to tighten the wiper arm on the shaft. Do not over-tighten.

3.7.2 Wiper motor replacement

Replace the wiper motor with a new one from the motor replacement kit (sold separately).

1. Remove the wiper from the sensor.
2. Use a #2 Phillips screwdriver to remove the three faceplate screws. Keep these screws to use again.
3. Lift the copper faceplate off the sensor.
4. Hold the motor shaft and pull straight up. It may be necessary to use pliers to carefully hold the shaft.

Figure 3 Wiper motor pulled from the sensor



5. Flush the wiper motor cartridge bore with fresh water two or three times.
6. Dry the wiper motor cartridge bore with canned compressed gas.
7. Flush the wiper motor cartridge bore with isopropyl alcohol to remove any remaining water.
8. Dry the wiper motor cartridge bore with canned compressed gas again.
9. Put the new wiper motor cartridge into the bore until it is flush with the optics face. Note that the wiper motor cartridge may not stay flush until the copper faceplate is attached again.
10. Align the dowel holes on the wiper motor cartridge and bore, and insert the plastic alignment dowel again.
11. Attach the faceplate to the optical face again with the faceplate screws.
12. Put the wiper on the motor shaft. Refer to [Wiper replacement](#) on page 13.

Figure 4 Copper faceplate attached to optical face



The faceplate fits on the optical face only one way.

3.7.3 Clean bulkhead connectors

NOTICE

Do not use WD-40® or petroleum-based lubricants on bulkhead connectors. It will cause damage to the rubber.

Damaged connectors can cause a loss of data and additional costs for service.

Damaged connectors can cause damage to the sensor and make it unserviceable.

Examine, clean, and lubricate bulkhead connectors at regular intervals. Connectors that are not lubricated increase the damage to the rubber that seals the connector contacts. The incorrect lubricant will cause the failure of the bulkhead connector.

1. Apply isopropyl alcohol (IPA) as a spray or with a nylon brush or lint-free swab or wipes to clean the contacts.
2. Flush with additional IPA.
3. Shake the socket ends and wipe the pins of the connectors to remove the IPA.
4. Blow air into the sockets and on the pins to make sure they are dry.
5. Use a flashlight and a magnifying glass to look for:

Operation and maintenance

Any corrosion.		
Cracks, scratches, or other damage on the rubber pins or in the sockets.		
Separation of the rubber from the pins.		
Swelled or bulging rubber pins.		

6. Use a silicone-based lubricant on each of the contacts of the bulkhead connector. The manufacturer recommends any of the products listed below.
 - 3M™ Spray Silicone Lubricant (3M ID# 62-4678-4930-3). Make sure to let it dry.
 - Dow Corning Molykote® III Compound (DC III)
 - Dow Corning High Vacuum Grease® (DC 976 V)
 - Dow Corning 4 Electrical Insulating Compound® (DC 4)
 - Dow Corning Molykote 44 High Temperature Grease® (DC 44)

Use a finger to put a small quantity (approximately 1 cm in diameter) of silicone grease on the socket end of the connector and push as much of the lubricant as possible into each socket. Do not use too much lubricant, as that will prevent a good seal.



7. Connect the connectors.
8. Use a lint-free wipe to clean any unwanted lubricant from the sides of the connectors.

Section 4 Reference

4.1 Delivered items

- the *ECO* sensor
- a dummy connector and lock collar
- blue-tipped power connector and lock collar for sensors with internal batteries
- a plastic protective cover for the optical face
- a model-specific spare parts kit
- **On the CD:**
- this user manual
- the software
- the device file or files for the sensor
- the characterization or calibration page for the sensor.

4.2 Calibration

The manufacturer calibrates all scattering sensors to make sure that the data that is collected meets the specifications for the sensor. This information is on the sensor-specific calibration page that comes with the sensor.

4.3 Characterization

The manufacturer uses a fluorescent material to characterize all fluorescence sensors to make sure that the data that is collected meets the specifications of the sensor. This information is on the sensor-specific characterization page that comes with the sensor.

4.3.1 Field characterization

The manufacturer recommends that the user perform a field characterization on fluorometers to make sure that the data is as accurate as possible for the user's application. The scale factor and the dark counts values can vary depending on the natural water, temperature, length of the cable, the power supply, and other factors.

Do the steps below to field-characterize the sensor.

- **x** = a solution of a known concentration in volts or counts.
 - **output** = the measured sample of interest in volts or counts.
 - **dark counts** = the measured signal output in volts or counts of the sensor in clean water with black tape over the detector.
 - **scale factor** = the multiplier in $\mu\text{g/L/volt}$, ppb/L/volt , OR $\mu\text{g/L/count}$, ppb/L/count .
1. Get a solution of a known concentration, **x**.
 2. Measure and record this solution using the sensor.
This value is the **output** in volts or in counts.
 3. Measure and record the sensor's **dark counts**.
 4. Use this equation to determine the sensor's **scale factor**:
$$\text{Scale factor} = x \div (\text{output} - \text{dark counts}).$$
 5. Use the scale factor to determine the concentration of the sample of interest:
$$(\text{output counts} - \text{dark counts}) \times \text{scale factor} = \text{concentration of solution}.$$
 6. Store the scale factor and dark counts (offset) in the device file for the sensor, the internal memory of the sensor, or both.

4.3.2 Store field characterization values in sensor

The host software uses a device file to process data. An example is shown below.

```

PCO BBFL2W-933
Created on: 01/22/12

Columns=9
N/U=1
N/U=2
N/U=3
Lambda=4    3.274e-06    50    700    700
N/U=5
chl=6        0.0121        51
N/U=7
cdom=8       0.091        49
N/U=9

```

1. Edit columns 6 and 8 of this device file to include the field characterization values.
2. Save this device file with a new name.
3. Open this device file in the host software under the *File* menu.

4.4 Device files

The software uses a sensor-specific device file to show the data in the *Plot Data* tab and to calculate the data output in engineering units. Each device file has three necessary elements. The device file is not necessary to use the software to set up and move data from a sensor.

1. The heading for the *Plot Data* tab.
2. The number of columns in the device file.
3. A description of the contents in each column.

Plot Data tab heading

The first line of the device file shows the model number and the serial number of the sensor. This information appears on the top of the *Plot Data* tab in the software.

Number of columns

The column count shows how many columns of data the software will process. The format is COLUMNS=x.

Description of columns

Each column of data from the sensor comes from a description in the device file.

Number of columns=x

Date=x MM/DD/YY

Time=x HH:MM:SS

N/U=x not used

sc=scale factor

off=offset

IENGR=x

mw=measurement wavelength of the sensor

dw=display wavelength of the sensor

Example of a three-parameter device file

lambda (scattering wavelength) = x sc off mw dw

Column 4 = scale factor (sc), offset (off),
measurement wavelength (mw),
display wavelength (dw).

Column 6 = scale factor (sc) offset (off).

Column 8 = scale factor (sc) offset (off).

```

PCO BBFL2W-933
Created on: 01/22/12

Columns=9
N/U=1
N/U=2
N/U=3
Lambda=4    3.274e-06    50    700    700
N/U=5
chl=6        0.0121        51
N/U=7
cdom=8       0.091        49
N/U=9

```

4.5 Terminal program operation

Use Windows HyperTerminal®, Tera Term, or other terminal program as an alternative to the manufacturer-supplied software to operate sensors.

baud rate: 19200	stop bits: 1	data bits: 8	flow control: none	parity: none
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4.5.1 Common terminal program commands

Command	Parameters	Description
!!!!	none	Stops the data collected by the sensor. Lets the user to enter setup values. If the sensor is in a low-power mode, turn the power supply off for one minute, then turn the power on and push the "!" key 5 or more times.
\$ave	1–255	The number of measurements that make up each row of collected data.
\$mnu	—	Prints the menu of setup values to the PC screen.
\$pkt	0–65535	Sets the number of rows of data that are collected between the specified time intervals.
\$rls	none	Gets the settings from the flash memory.
\$run	—	Uses the current setup values to operate.
\$sto	—	Saves the desired setup values to the flash memory.

4.5.2 Terminal program commands for sensors with internal memory

Command	Parameters	Description
\$clk	24-hour time	Sets the time in the internal memory in the format hhmmss.
\$date	date	Sets the date in the internal memory in the format mmddyy.
\$emc	—	Clears the internal memory.
\$get	—	Reads data from the internal memory. Prints etx when it is complete.
\$int	24-hour time	Sets the time interval between sets of measurements the format hhmmss.
\$mvs	1 = ON; 0 = OFF	1 = the Bio-wiper is open. 0 = the Bio-wiper is closed.
\$rec	1 = ON 0 = OFF	1 = Turns on the sensor's internal memory. 0 = Turns off the sensor's internal memory.
\$rls	—	Loads the settings from the flash memory.
\$set	0–65535	Sets the number of rows of data that are output between low-power states.

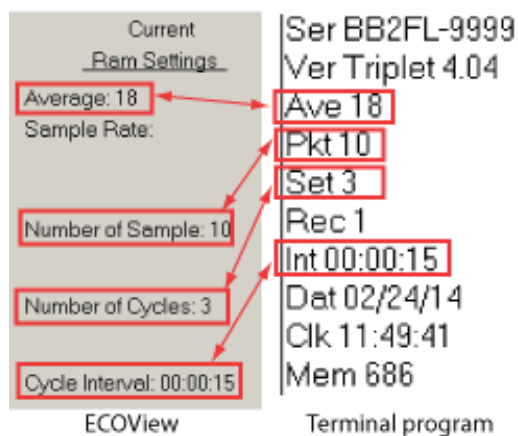
\$m1d	0 to 65535	Parameter 1, dark count value to calculate the data collected in engineering units.
\$m1s	float	Parameter 1, slope value to calculate the data collected in engineering units.
\$m2d	0 to 65535	Parameter 2, dark count value to calculate the data collected in engineering units.
\$m2s	float	Parameter 2, slope value to calculate the data collected in engineering units.
\$m3d	0 to 65535	Parameter 3, dark count value to calculate the data collected in engineering units.
\$m3s	float	Parameter 3, slope value to calculate the data collected in engineering units.
\$man	1 = enable 0 = disable	Enables or disables manual start time. The sensor starts at a programmed wake time when enabled. This is automatically enabled if the user enters a manual start time.
\$mst	24-hour time	If enabled, the sensor will start at a pre-programmed time. This allows all sensors on a system to be synchronized to collect data.
\$met	—	Prints metadata that describes the data that is collected. Field 1 is the data. Field 2 is the field tag. Field 3 is the field name. Field 4 is used for units where appropriate.

Reference

\$rat	2400 to 230400	Baud rate for communication. An invalid baud rate will default to 19200. It will show as 19201 so the user knows the software defaulted to this value.
\$rfd	—	Loads the original manufacturer's setup.
\$seq	0 to 3	Selects one of the pre-defined sequences to use to collect data.

4.5.3 ECOView and terminal program differences

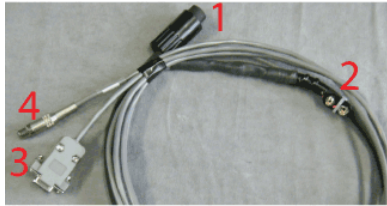
ECOView uses different but equivalent words for data collection options.



Section 5 Optional equipment

5.1 Test cable

Use a test cable to set up and test the sensor before deployment.



One analog output connector



Two analog output connectors



No analog output

1 six-contact connector	3 db-9 serial port connector
2 9-volt battery connector	4 RCA connector(s)

1. Connect the six-contact connector to the sensor.
2. Connect the 9-volt connector to a 9-volt battery. As an alternative, it can be connected to a regulated power supply.
3. Connect the db-9 connector to the PC. Use a USB-to-RS232 adapter cable if necessary.
4. Use a digital multimeter (DMM) to see analog output (if applicable) from the sensor. The inside of the RCA is signal (red DMM probe) and the outside is ground (black DMM probe).

5.2 Internal batteries

⚠ WARNING

The pressure housing of the *ECO* sensor must be opened to replace the batteries. If done improperly, it could result in personal injury or death due to abnormal internal pressure as a result of flooding. It may not be possible to repair flooded sensors.

The manufacturer disclaims all product liability from the use or service of these sensors. The manufacturer cannot control the use of these sensors or choose qualified personnel to operate them, and therefore cannot take steps to comply with laws pertaining to product liability, including laws that impose a duty to warn the user of any dangers involved with the operation and maintenance of the sensors. Acceptance of these sensors by the customer shall be conclusively deemed to include a covenant by the customer to defend and hold the manufacturer harmless from all product liability claims arising from the use and servicing of these sensors. Flooded sensors are serviced at the discretion of the manufacturer.

⚠ WARNING

The sensor may be under pressure. Do not point it toward any part of the body to remove the vent plug or end flange.

⚠ WARNING

Replace the batteries in a clean and dry environment. Gases in the sensor can expand and push the pressure relief plug open. This will flood the sensor. Do not replace batteries in a cold environment and then deploy the sensor in a hot environment.

Flooded sensors are usually not serviceable. There is a small chance that the manufacturer can get the data that is stored in the sensor. Contact service@wetlabs.com for more information about flooded sensors.

The manufacturer supplies six Ultralife® 9-volt lithium batteries to supply power to sensors that have internal batteries. The manufacturer recommends these batteries because six batteries supply approximately 6100 mAh, which is the highest capacity currently available.

The use of alkaline or lithium manganese dioxide (LiMnO₂) batteries will not damage the sensor but the capacity is much lower than UltraLife® batteries.

Note: The nominal water temperature, sequence timing, sample periods, and other variables will change the use-time of the sensor's internal batteries.

5.2.1 Remove batteries

1. Clean any debris from the end flange.
2. Dry the sensor thoroughly.
3. Remove the dummy plugs if necessary.
4. Point the connector end flange down, away from the face.
 - a. Pull the vent plug loose.
 - b. If the sensor has an external thermistor, pull it loose.
5. Dry the vent plug (and the thermistor, if applicable).
6. Use needle-nose pliers to remove the monofilament from the end flange.

Figure 5 Pull monofilament from end flange



7. Remove the end flange from the pressure housing.

The jacking screws supplied as spare parts can be used to push the end flange off of the pressure housing, and then be removed.
8. Carefully disconnect each Molex® connector.
9. Remove the screw that holds the vent plug in the end flange.
10. Dry the end flange and the pressure housing seal areas.
11. Examine the O-rings on the vent plug and thermistor (if applicable).

Remove any O-ring that is damaged.
12. Apply a light coat of vacuum grease on a new O10 O-ring and put it on the vent plug or thermistor.
13. Put the vent plug into the top of the end flange.
14. If applicable, insert the thermistor into the end flange again.
15. Put the screw for the vent plug in the inside of the end flange.

The screw holds the vent plug in the end flange.
16. Carefully pull the white plastic loop to remove the battery pack from the pressure housing.
17. Remove the black plastic protectors from the ends of the long screws that secure the batteries.
18. Loosen, but do **not** remove, the retaining screws using a 1/4" slotted driver.

Do not remove both screws. If both of the retaining screws are removed, the result will be a pile of parts and the replacement process will not be easy.

Figure 6 Both retaining screws removed



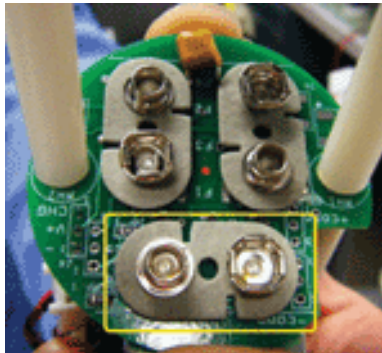
19. Disconnect each of the six batteries.

5.2.2 Replace batteries

Install new batteries in the sensor.

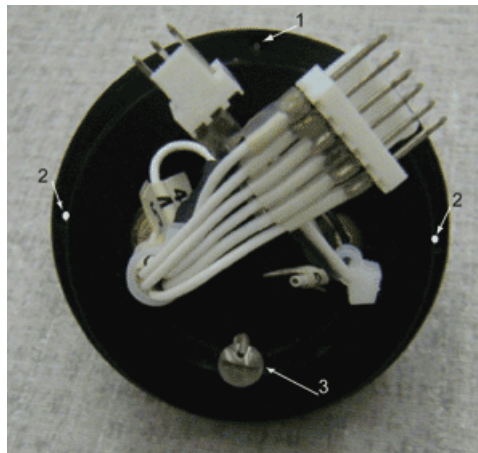
1. Tilt the battery board enough to connect the first battery in the contacts that are perpendicular to the other two.

Figure 7 Connect the first battery



2. Pivot the boards in the opposite direction to connect the other two batteries.
3. Connect the second set of the batteries.
4. Hold the assembly plates on top and bottom and tighten the screws.
The bottoms of the batteries may splay out. Make sure that the batteries do not extend beyond the circuit boards. If they do, the battery pack will scratch the sealing surface when it is put into the pressure housing again.
5. Install the bottom neoprene pad and the black plastic protective covers on the ends of the screws.
6. Remove and examine the 224 O-ring on the pressure housing for damage.
7. Use a new O-ring if necessary.
8. Apply a light layer of grease such as Dow Corning® High Vacuum Grease on the O-ring.
9. Put the battery pack into the pressure housing.
10. Attach the Molex® connectors.
Note that the dowel pin may be in the end flange or in the pressure housing.

Figure 8 Inside of end flange



1 dowel hole

2 through-holes for jacking screws

3 vent plug screw

11. Put the end flange onto the housing so that the wires are out of the way.
12. Align the dowel pin with the dowel hole in the end flange, not the jack screw holes, which go through the end flange.
13. Make sure that the wires are not pinched between the end flange and the pressure housing.
14. Make sure that the end flange is attached to the pressure housing.
15. Install the monofilament in the end flange.

5.3 Discontinued equipment: pressure sensor

This optional equipment is no longer available on new ECO sensors as of 15 June 2015.

The calibration coefficient value for the pressure sensor is on the calibration page that ships with the sensor. The pressure sensor gives pressure output in counts. Do one of the methods below to change counts to engineering units.

1. Use the drop-down menu in the *Plot Data* tab of the software to see the pressure sensor output in dbar.
2. Use MATLAB, MS Excel or other software to solve for:
relative pressure, dbar = (output × slope) + intercept
3. Calculate the absolute pressure:
 - absolute pressure, dbar = relative pressure, dbar - relative pressure at atmospheric-water interface, dbar.

Set the pressure sensor to zero for each deployment. Do not deploy the pressure sensor deeper than the depth rating on the calibration page.

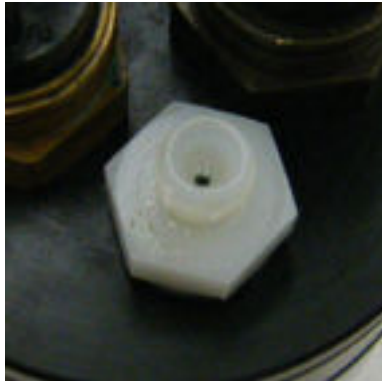
Pressure sensor maintenance

The plastic fitting filled with silicone oil is a buffer between the pressure transducer diaphragm and seawater. Add silicone oil to the reservoir on top of the transducer at regular intervals.

1. Make sure the top of the sensor is clean.
2. Use a 9/16" wrench to hold the white nylon Swagelok® fitting.
3. Use a 7/16" wrench to loosen the cap on top of the fitting.
4. Remove the cap.
5. Use a wire or toothpick to clean the hole in the cap.

- Do not blow compressed air into the fitting. It will make a mess.
6. Add Dow Corning® 200 Silicone Oil to the reservoir until the oil is in view.

Figure 9 Pressure sensor cap



7. Replace the cap. Make sure not to tighten the cap too much.
8. Clean any excess oil from the end flange of the sensor.

5.4 Discontinued equipment: external thermistor

This optional equipment is no longer available on new *ECO* sensors as of 15 June 2015.

The calibration coefficient value for the thermistor is on the sensor's characterization page that ships with the sensor. The thermistor gives temperature output in counts. Do one of the methods below to change counts to engineering units.

1. Use the drop-down menu in the *Plot Data* tab of the software to see the thermistor output in °C.
2. Use MATLAB, MS Excel or other software to solve for:
 - Temperature, °C = (Output × Slope) + Intercept

Section 6 General information

WARNING

This product can expose the user to chemicals with silica, crystalline (airborne particles of respirable size), which is known to the State of California to cause cancer and birth defects or other reproductive harm. For more information, go to www.P65Warnings.ca.gov.

Revised editions of this user manual are on the manufacturer's website.

6.1 Warranty

Refer to the manufacturer's website for warranty information (seabird.com/warranty).

6.2 Service and support

The manufacturer recommends that sensors be sent back to the manufacturer annually to be cleaned, calibrated, and for standard maintenance.

Refer to the website for FAQs and technical notes, or contact the manufacturer for support at support@seabird.com. Do the steps below to send a sensor back to the manufacturer.

1. Complete the online Return Merchandise Authorization (RMA) form or contact the manufacturer.
Note: *The manufacturer is not responsible for damage to the sensor during return shipment.*
2. Remove all batteries from the sensor, if so equipped.
3. Remove all anti-fouling treatments and devices.
Note: *The manufacturer will not accept sensors that have been treated with anti-fouling compounds for service or repair. This includes AF 24173 devices, tri-butyl tin, marine anti-fouling paint, ablative coatings, etc.*
4. Use the sensor's original ruggedized shipping case to send the sensor back to the manufacturer.
5. Write the RMA number on the outside of the shipping case and on the packing list.
6. Use 3rd-day air to ship the sensor back to the manufacturer. Do not use ground shipping.
7. The manufacturer will supply all replacement parts and labor and pay to send the sensor back to the user via 3rd-day air shipping.

6.3 China RoHS disclosure table

Name of Part	Hazardous substance or element in product					
	Pb	Hg	Cd	Cr(VI))	PBB	PBDE
PCBs	X	O	O	O	O	O
Battery pack	X	O	O	O	O	O
Cables	X	O	O	O	O	O
Housing	O	O	O	O	O	O
Mounting hardware	O	O	O	O	O	O
Accessories	O	O	O	O	O	O
This table is compiled to the SJ/T 11364 standard.						
O: This hazardous substance is below the specified limits as described in GB/T 26572. X: This hazardous substance is above the specified limits as described in GB/T 26572.						

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